Chain Link Security: Trends in Simultaneous HTTPS, ECH, and DNSSEC Use

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Goal & Motivation

- Usage of DNS HTTPS records benefits greatly from the simultaneous deployment of ECH and DNSSEC
  - ECH allows for the encryption of TLS handshakes, DNSSEC is a series of cryptographic signature protocols that ensure better integrity and authenticity of DNS records.
- It's important to use DNSSEC in conjunction with HTTPS to authenticate HTTPS records and prevent forgery by attackers, and to use it with ECH to properly authenticate handshake keys.
- What will be the larger environment for deploying DNS HTTPS records with TLS-encrypted ClientHello (ECH) and DNSSEC by 2025?
  - Using the data of 2025's popular domains as our subject, what is the larger environment for deploying DNS HTTPS records with TLS-encrypted ClientHello (ECH) and DNSSEC in 2026?
- In terms of deployment volume, what changes have occurred in the usage trends of HTTPS, ECH, and DNSSEC?

Background - HTTPS & TLS

- HTTPS uses TLS provide data confidentiality or authentication between the client and the server for services that require them
- TLS is the cryptographic protocol that provides encryption and authentication for HTTPS
- TLS negotiation begins with a handshake between the client and the server, during which cryptographic keys are generated to establish a secure channel for data transmission, and all data is then encrypted using those keys

Background - HTTPS Parameters & Dynamic Configs

- SvcPriority: The priority of the record
  - 0: AliasMode, which alias a domain to the target domain
  - Other values: ServiceMode, which provides information specific to a service endpoint
- TargetName: a domain name that can be either the alias target in AliasMode or the alternative endpoint in ServiceMode
- SvcParams
  - alpn (Application-Layer Protocol Negotiation): specifies additional application protocols
  - ipv4hint/ipv6hint: suggest IPv4/IPv6 addresses for reaching the endpoint
- Dynamic Configurations
  - Occurs when at least one parameter uses an expression instead of a fixed value
  - <ftp:config name="ftp" **defaultWriteEncoding="#[attributes.queryParams.encoding]**">, <ftp:connection username="foo" password="mySecret" />, </ftp:connection>

Background - ECH

- An extension of TLS that hides the identity of the target TLS server by encrypting the Server Name Indication (SNI) in the client Hello using the public key of the client-facing server
- Limits access to information that a particular user is visiting your website, ensuring that it is not unnecessarily shared with intermediaries
- Though it provides some mitigations, ECH's coverage of traffic analysis is incomplete
    - Traffic analysis involves collecting and analyzing the properties of communication channels that can leak portions of the ciphertext without breaking the underlying encryption scheme

Background - DNSSEC

- A suite of specifications designed to fortify DNS data with origin authentication, data integrity, and authenticated denial of existence
- Is used to prevent DNS spoofing attacks by allowing authoritative name servers to digitally sign RRsets using public-key cryptography
- DNSSEC process: a DNSSEC resolver verifies that the information originates from a legitimate server and that the data matches the information on authoritative DNS servers by checking signatures in the DNSSEC servers' answers
- Potential Vulnerability: It is unable to guard against poor configuration or bad information in the authoritative name server, and defend against buffer overruns or DDoS attacks

Approach

- We would query the top 10,000 pay-level (no sub-domains) domains from over the course of 2025, as well as for each month of the year, using a list provided by the site Tranco.
- We fetch the information regarding the following Parameters:
  - HTTPS RR (Does the domain use HTTPS)
  - ECH Usage (Does the domain use ECH)
  - DNSSEC Usage (Does the Domain use DNSSEC)
  - SvcPriority Mode Used (Service or Alias Mode)
  - ALPN Usage
  - ipv4hint usage
  - Ipve6 usage
  - Dynamic Configuration usage
- We extracted and calculated the data of each list, but notably we only had the resources to extract the information of the date of extraction, not the HTTPS protocol use of that specific period.
  - Our finalized extraction of data occurred on March 7th, 2026.

Approach Continued

- We queried each domain using the library DNSPython to ask them what their HTTPS Resource Record said.
  - Specifically, we asked for each snippet of data, parse that data by checking for certain keywords, numbers, or bits within sections of the record. If the parsed data returns a hit, then we tally up our count of that particular parameter.
- Additionally, we take that tally and calculate overall averages, then multiply that by a hundred to get our percentages.
  - These percentages are rounded to the second decimal.
- The Tranco lists we used applied Harmonic progression/Dowdall rule to a set of source lists.
  - Chrome User Experience Report (only applicable to Chrome traffic)
  - Majestic
  - Cloudflare Radar (does <u>not</u> only cover Cloudflare domains)
  - Cisco Umbrella (based on DNS traffic to Cisco Umbrella/OpenDNS resolvers)

Ethics

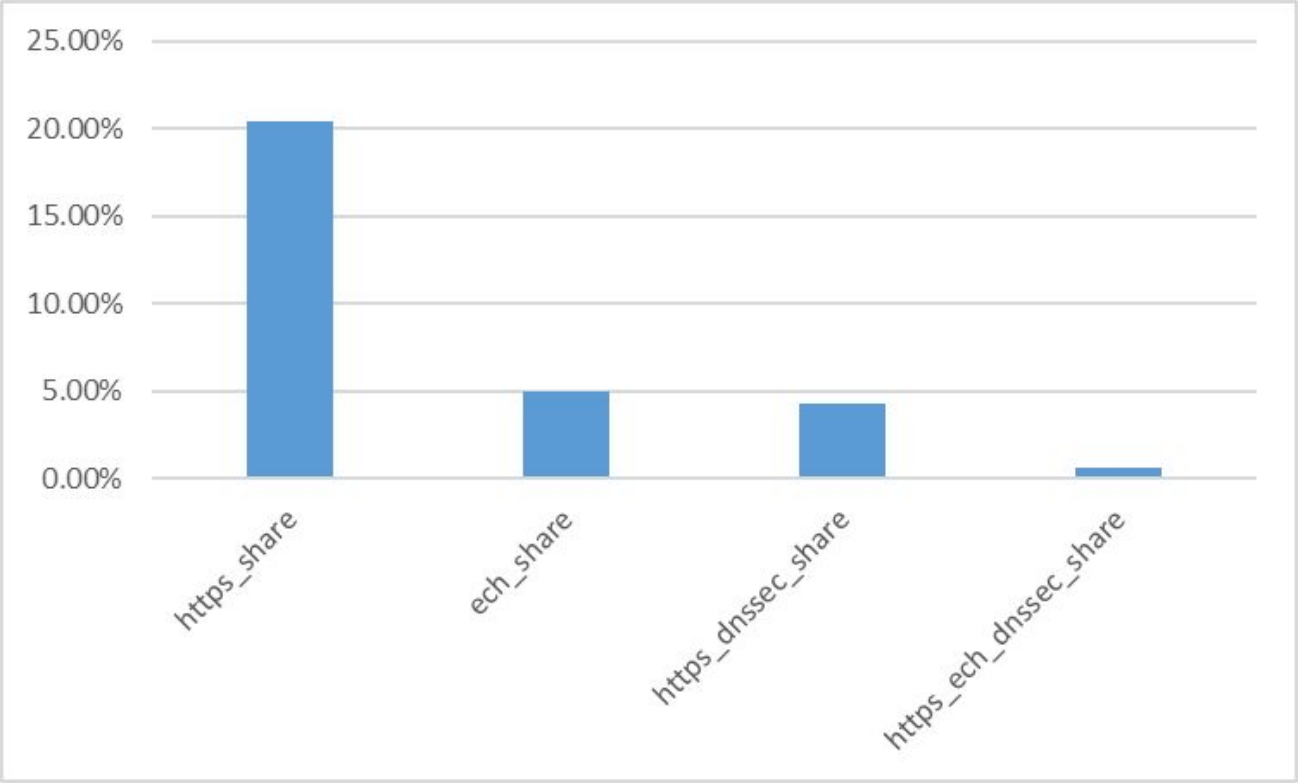
- To ensure our data collection and measurement are both safe and ethical, we need to exercise caution when collecting data, avoid overloading DNS, and adhere to all relevant data-collection consent standards.
- Minimize the workload on the domains we're fetching
  - Asking for measured parameters and usage details only from the domains without the extra data we don't need
  - Sending too many requests to a domain can cause server overload, bandwidth consumption, or exhaustion of computing resources
- Respond to any requests to stop measuring the data

Result

Domain/Protocol	HTTPS	ECH	DNSSEC	HTTPS+ECH+DNSSEC
Count	2044	497	430	59
Share(%)	20.44	4.97	4.3	0.59

Domain/HTTPS params & dynamic config	Alias mode	Service mode	alpn	ivp4	ivp6	Dynamic config
Count	3	2041	2029	1977	1689	2037
Share(%) vs all domain	0.03	20.41	20.29	19.77	16.89	20.37
Share(%) vs https domain	0.14	99.85	99.27	96.72	82.63	99.66

Simultaneous HTTPS, ECH, and DNSSEC usage

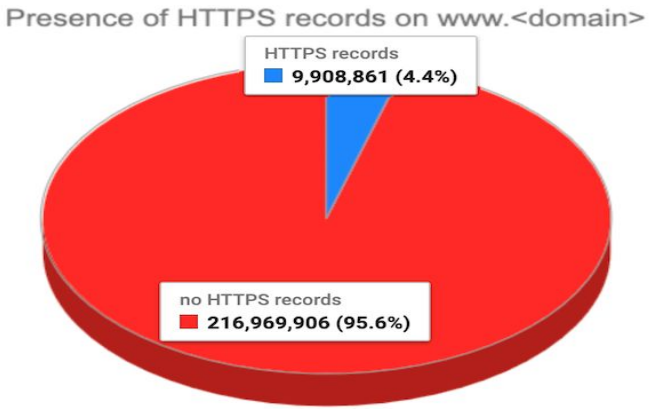


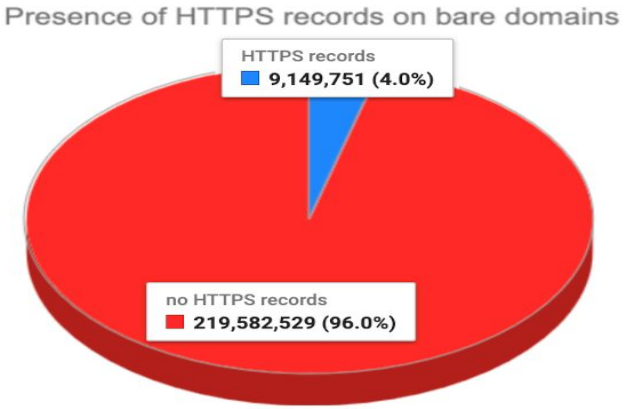
HTTPS Adoption

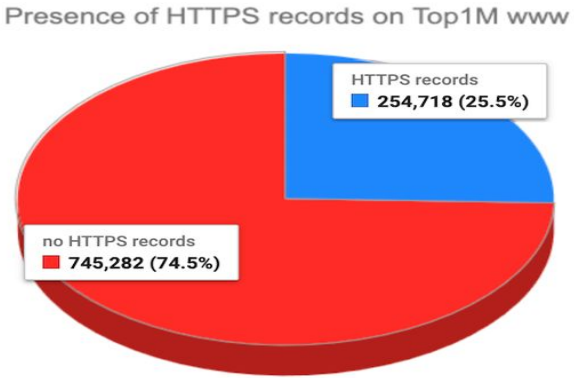
- 2044 out of 10,000, or 20.44% in the 2026 environment.
- Roughly pretty similar to the 20% to 27% range of the top one million from June 2023 to April 2024 in the previous study [1].
- Look up the roadblocks to HTTPS adoption to explain why it’s largely static.

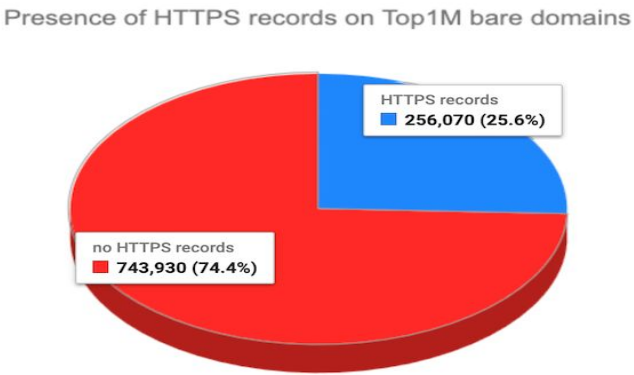
With HTTPS, data is scrambled when it is sent over the web and decoded after it reaches its intended destination. Because of the nature of the encrypted transmission, even if someone were to intercept the data, they wouldn’t be able to read it.

Presence of HTTPS records









Comparing the Results to Existing Work

- Notable trends saw a decrease in simultaneous HTTPS and DNSSEC usage decrease significantly from the previous study (Exploring the Ecosystem of DNS HTTPS Resource Records: An End-to-End Perspective) [1]
- The top 10,000 domains of 2025 overall had a DNSSEC presence of 4.30%, or 430 domains out of 10,000. This was done counting either the presence of the AD bit or RRSIG as a single tally.
- For reference, the 2024 paper had a RRSIG presence of range of 13% to 16% of one million domains across the dates of June 2023 to April 2024.
    - They do not measure the presence of the AD bit by itself.

DNSSEC adoption overview

- DNSSEC adoption remains relatively low across the global internet.
- According to whisper security, out of 209 million domains they analyzed, only about 4.7% use DNSSEC.
- According to CircleID, as of 2023, approximately 92% of top-level domains (TLDs) are DNSSEC-signed. However, these numbers drop significantly as you go down domain levels.
- In 2024, only about 5% of .com domains used DNSSEC.
- Adoption varies significantly depending on governance and policy.
- Security deployment is more common among popular websites.

Observation:

- Top 1 million websites are 2.7× more likely to use DNSSEC compared to the average domain
- The infrastructure is there, but why has actual implementation taken a hit?

DNSSEC vs HTTPS

Common Misconception:

- The key argument made is that HTTPS makes DNSSEC redundant.

What actually happens:

- HTTPS only secures the connection after DNS query resolution. This means it doesn’t stop an attacker from redirecting the query.
- DNSSEC protects the DNS lookup process itself and remains the only standardized way of doing so.

Threats DNSSEC Helps Prevent

- DNS spoofing
- Cache poisoning
- Redirection attacks

Benefits vs Challenges

Benefits of DNSSEC	Challenges of DNSSEC implementation
Data integrity	Increased computational load
Authentication	New security concerns
User trust is increased	Increased complexity
Business safety and continuity	Additional maintenance requirements
	Cost and increased overhead

ECH usage

- Slightly fewer than 10% of the top million websites (by traffic) support ECH, and just 0.06% of connections actually use ECH. This is due to a combination of device-side and server-side factors
- From the client side, users need a browser that supports ECH — Chrome and Firefox, for example — and to configure the system to use encrypted DNS to hide their DNS queries and access a compatible resolver.
- On the server side, the simplest way for a website to enable ECH is to work with a content delivery network (CDN) that supports it. Today, the only major CDN that supports ECH is Cloudflare
- Because ECH requires coordinated support across DNS, TLS, and server infrastructure, adoption remains significantly lower than HTTPS resource record adoption.

HTTPS + ECH

- 497 domains out of 10,000, or 4.97% in 2026.
- Notable because at the time of writing in the 2024 paper, ECH had been depreciated in Cloudflare domains after October 5th, 2023.
  - Up to that point

Simultaneous Adoption of HTTPS + ECH + DNSSEC

- Only 0.59%, or a measly 59 out of 10,000.
- Notably not a statistic covered by the 2024 paper [1]
- Possible reasons why it’s so low:
  - I have some analysis ideas that involve examining the raw data, but I didn’t want to scare you guys with another big seeming task (it’s not that bad), so I’ll tell you guys in class lol - Alex

Conclusion

Citations

[1]Hongying Dong, Yizhe Zhang, Hyeonmin Lee, Shumon Huque, and Yixin Sun. 2024. Exploring the Ecosystem of DNS HTTPS Resource Records: An End-to-End Perspective. In Proceedings of the 2024 ACM on Internet Measurement Conference (IMC '24). Association for Computing Machinery, New York, NY, USA, 423–440. https://doi-org.oregonstate.idm.oclc.org/10.1145/3646547.3688410

[2] J. Zirngibl, P. Sattler and G. Carle, "A First Look at SVCB and HTTPS DNS Resource Records in the Wild," 2023 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW), Delft, Netherlands, 2023, pp. 470-474, doi: 10.1109/EuroSPW59978.2023.00058.

[3] Prob the Tranco lists, this is a placeholder

Questions?

Thank You!